The Bus Depot Is a Virtual Power Plant

26 utilities are turning idle school buses into grid batteries — routes, health, tariffs and grid dispatched together.

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Guide: Dr. Neel Kapoor · Smart Mobility Lab

Electrical Engineering · Final year

Parked Buses Can Carry the Peak

Controls make it dispatchable, operations keep routes covered, economics make it pay.

01

Grid Control Stack

Managed charging, bidirectional flow, and V2G dispatch to cut peaker demand

02

Fleet Operations

Charge around routes and hold reserve for shelters and outage backup

03

Economics and Pilot

New rates plus pilots that sent 27.7 MWh back over 316 hours

01

GRID CONTROL STACK

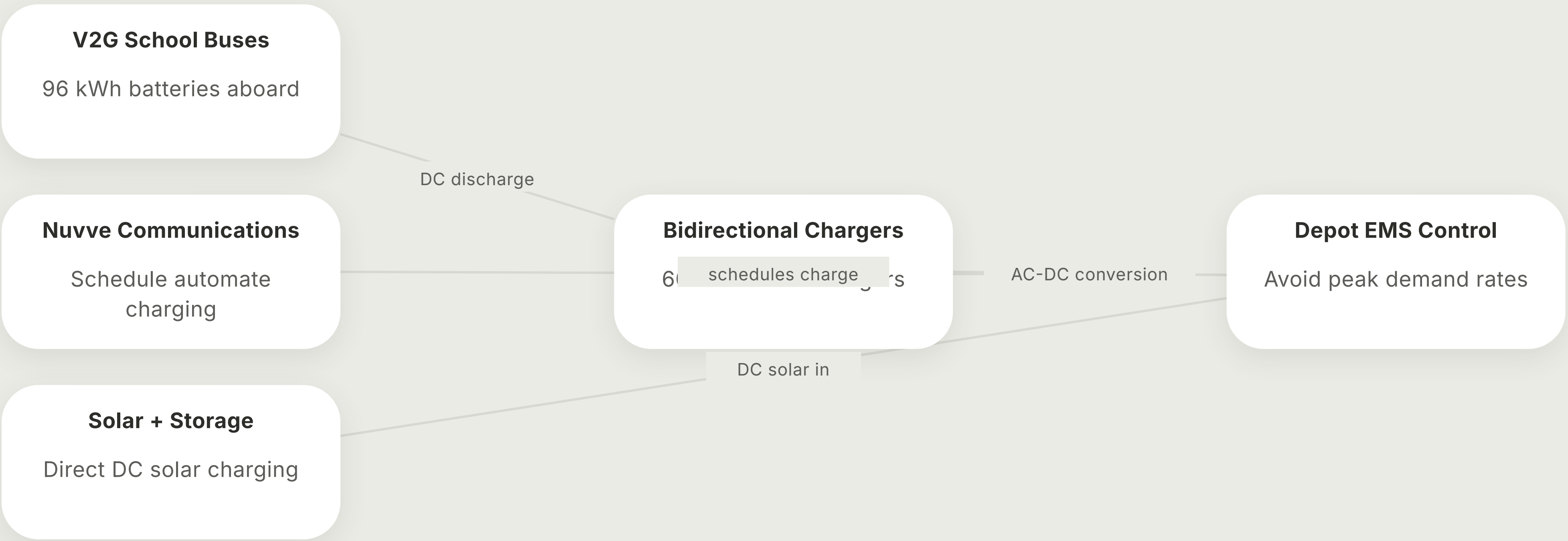
Four Controls Turn Parked Buses Into Dispatchable Power

BMS, charger, depot EMS, and utility DERMS act as one stack proven over 50 summer hours.

Utility DERMS Link	Dispatch to grid	50+ hrs proof · 5-bus pilot
Depot EMS Control	Schedules charging	Automate · Avoid peaks
Bidirectional Charger	12.5kW to 25kW	DC power bus · V2G + V2H

60kW Chargers Only Matter When Wired Into One VPP Path

Buses discharge through bidirectional chargers under EMS control to feed the grid at peak demand.



No Bus Exports Until Its Morning Route Is Safe

Grid request comes last — route, battery, then price decide if a parked bus answers.

1. Reserve tomorrow's route

Hold energy for the next run first. No export leaves a bus short in the morning.

2. Prove battery can give

Check charge level and bidirectional link. Charge and discharge paths must be ready.

3. Check price beats peak

Confirm peak export beats charge cost. Managed charging avoids peak demand rates.

4. Answer grid for pay

Only then send energy back to the grid and earn compensation for the depot.

02

Student Routes Run First — The Grid Gets the Idle Hours

Morning and afternoon runs are untouchable — buses earn grid value only in the parked window between.

Routes First, Then Buses Carry the Peak

Set schedules create set idle windows: midday park, evening export, overnight recharge.

6 AM	Morning routes run; no export until buses return and next-run energy is safe.
9 AM	Midday idle begins — buses sit parked while daytime electricity demand climbs.
2 PM	Afternoon routes run; buses return and reserve tomorrow's mileage first.
5 PM	Peak discharge: send stored power when demand is highest and most expensive.
11 PM	Off-peak recharge overnight on time-of-use rates for the next school day.
Summer	Long idle unlocks scale: Beverly sent 27.7 MWh back over 316 hours.

V2G fails unless five links lock in order

Massachusetts proved it in 2015: one missing link stalls all export.

Morning routes locked

Routes first, idle to grid

Trained staff ready

Lack of staff blocked V2G test

Bidirectional chargers

DC chargers enable export

Interconnection cleared

Fast-track under-resourced sites

V2G rate structure

Avoid 15-min demand charges

Peak dispatch pays

50+ hours cut peaker need

Routes Stay On Time While Buses Carry the Peak

One board holds service, availability and battery reserve beside proven export hours and MWh.

On-time routes

First

—

Buses ready

5

—

Peak hours cut

316

▲ 27.7MWh

MWh to grid

31.7

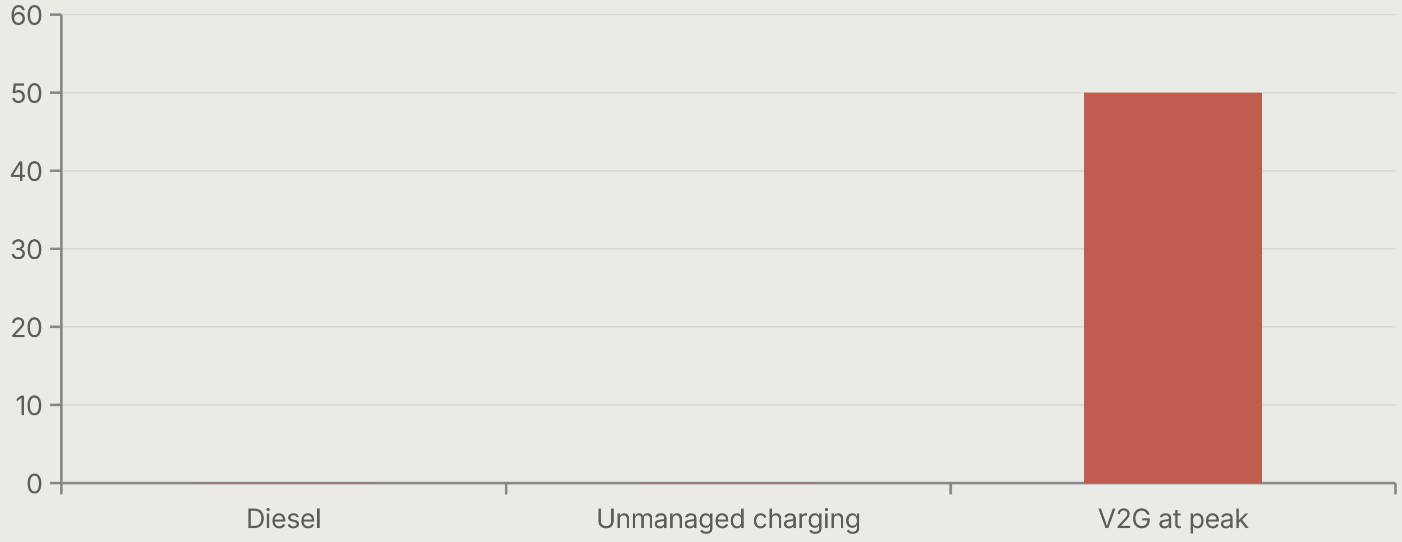
▲ 282 h

03

Translate V2G capability into a bankable, equitable depot plan.

Unmanaged Charging Spikes Bills — V2G Turns Peak Into Revenue

Diesel can't export; unmanaged charging risks 15-minute demand charges, while V2G discharged 50+ hours at peak.



- Unmanaged peak charging can trigger hourly or 15-min demand charges.
- V2G shaves peak, avoids firing costly fossil peaker plants.
- One bus exported 50+ hrs in Beverly summer pilot.

Say it plain: only the V2G bar earns — the other two only cost.

Peak-Shaving Wins Paid Dispatch; Backup Wins Battery Life

UD earned roughly $1,200 per BEV yearly in PJM — scored here on money versus wear.

	Peak-Shaving	Resilience Backup	TOU Arbitrage
Revenue upside (w 0.5)	5	2	3
Battery friendly (w 0.3)	3	5	2
Simple to run (w 0.2)	3	4	2
Total	4	3.3	2.5

Pilot in Three Moves: Test One, Pay Five, Scale Fifty

Start with a single-bus discharge, then earn peak revenue before scaling an equity-first VPP.

	Months 0-3	Months 4-12	Year 2 onward
Phase 1: Prove	**Single-bus discharge** Replicate PEPCO fall 2024 first test		**Reserve morning route** No export until routes are safe
Phase 2: Earn Peak	**Deploy 5-bus peak cut** Repeat White Plains 2018-2021 test		**Export 27.7 MWh** Match Beverly: 316 hrs, 3 buses
Phase 3: Scale VPP	**Grow to 4-bus VPP** Add 31.7 MWh over 282 hours	**Prioritize low-income** 50 buses in marginalized routes	**Build resilience hub** Portland low-income community hub

Sources: Pilot Census, Measured Exports, Federal Playbooks

The depot-as-plant record — where buses discharged, for how long, and what rules paid them.

WRI Electric School Bus Initiative (Apr 2025) — Map + Table of Utility V2G ESB Pilots in 19 states.

Census behind pilot scale: fleet counts, durations, utilities and partners.

National Grid, Beverly MA, summer 2021 — one ESB exported 50+ hours during peak demand.

Early proof that peak discharge cut peaker-plant need and local emissions.

Highland + National Grid, Beverly 2021-2024 — 3 buses sent 27.7 MWh back over 316 hours.

Commercial follow-on quantifying multi-summer export volume and hours.

Con Edison + White Plains SD, 2018-Sept 2021 — 5 Lion/Nuvve buses in peak-shaving V2G test.

NY first: buses charged/discharged power distributed direct to customers.

Highland + Green Mountain Power, 2023-2024 — 4 buses sent 31.7 MWh back over 282 hours.

VPP-as-DER case: utility dispatched buses to optimize the grid.

Dispatch Routes First, Rates Second, Grid Third